# SNx4ACT241 Octal Buffers/Drivers with 3-State Outputs

## 1 Features

- Operation of 4.5V to 5.5V $V_{CC}$
- Inputs accept voltages to 5.5V
- Max $t_{pd}$ of 8.5ns at 5V
- Inputs are TTL-compatible

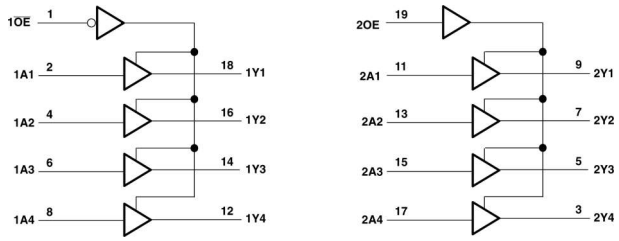
## 2 Description

These octal buffers and line drivers are designed specifically to improve the performance and density of 3-state memory address drivers, clock drivers, and bus-oriented receivers and transmitters.

**Device Information**

| PART NUMBER | PACKAGE(1) | PACKAGE SIZE(2) | BODY SIZE(3) |
|---|---|---|---|
| SNx4ACT241 | N (PDIP, 20) | 24.33mm × 9.4mm | 24.33mm × 6.35mm |
| | DW (SOIC, 20) | 12.8mm × 10.3mm | 12.8mm × 7.5mm |
| | NS (SOP, 20) | 12.6mm × 7.8mm | 12.6mm × 5.3mm |
| | DB (SSOP, 20) | 7.2mm × 7.8mm | 7.2mm × 5.3mm |
| | PW (TSSOP, 20) | 6.5mm × 6.4mm | 6.5mm × 4.4mm |

(1) For more information, see Section 10.
(2) The package size (length × width) is a nominal value and includes pins, where applicable.
(3) The body size (length × width) is a nominal value and does not include pins.




**Logic Diagram (Positive Logic)**

An IMPORTANT NOTICE at the end of this data sheet addresses availability, warranty, changes, use in safety-critical applications, intellectual property matters and other important disclaimers. UNLESS OTHERWISE NOTED, this document contains PRODUCTION DATA.

# Table of Contents

**1 Features**....1
**2 Description**....1
**3 Pin Configuration and Functions**....3
**4 Specifications**....4
4.1 Absolute Maximum Ratings....4
4.2 Recommended Operating Conditions....4
4.3 Thermal Information....4
4.4 Electrical Characteristics....5
4.5 Switching Characteristics....5
4.6 Operating Characteristics....5
**5 Parameter Measurement Information**....6
**6 Detailed Description**....7
6.1 Overview....7
6.2 Functional Block Diagram....7
6.3 Device Functional Modes....7
**7 Application and Implementation**....8
7.1 Power Supply Recommendations....8
7.2 Layout....8
**8 Device and Documentation Support**....9
8.1 Documentation Support....9
8.2 Receiving Notification of Documentation Updates....9
8.3 Support Resources....9
8.4 Trademarks....9
8.5 Electrostatic Discharge Caution....9
8.6 Glossary....9
**9 Revision History**....9
**10 Mechanical, Packaging, and Orderable Information**....10


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## 3 Pin Configuration and Functions

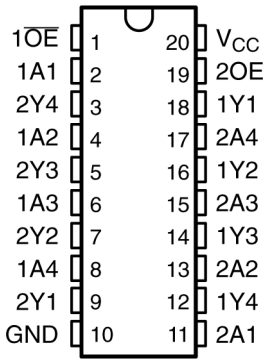



**Figure 3-1. SN54ACT241 J or W Package SN74ACT241 DB, DW, N, NS, or PW Package (Top View)**

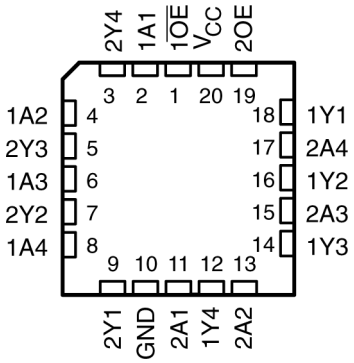



**Figure 3-2. SN54ACT241 FK Package (Top View)**

**Table 3-1. Pin Functions**

| NAME[1] | PIN | TYPE | DESCRIPTION |
|---|---|---|---|
| 1OE | 1 | I | Output enable 1 |
| 1A1 | 2 | I | 1A1 input |
| 2Y4 | 3 | O | 2Y4 output |
| 1A2 | 4 | I | 1A2 input |
| 2Y3 | 5 | O | 2Y3 output |
| 1A3 | 6 | I | 1A3 input |
| 2Y2 | 7 | O | 2Y2 output |
| 1A4 | 8 | I | 1A4 input |
| 2Y1 | 9 | O | 2Y1 output |
| GND | 10 | — | Ground pin |
| 2A1 | 11 | I | 2A1 input |
| 1Y4 | 12 | O | 1Y4 output |
| 2A2 | 13 | I | 2A2 input |
| 1Y3 | 14 | O | 1Y3 output |
| 2A3 | 15 | I | 2A3 input |
| 1Y2 | 16 | O | 1Y2 output |
| 2A4 | 17 | I | 2A4 input |
| 1Y1 | 18 | O | 1Y1 output |
| 2OE | 19 | I | Output enable 2 |
| VCC | 20 | — | Power pin |

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# 4 Specifications

## 4.1 Absolute Maximum Ratings

over operating free-air temperature range (unless otherwise noted)(1)

| | | | MIN | MAX | UNIT |
|---|---|---|---|---|---|
| $V_{CC}$ | Supply voltage range | | –0.5 | 7 | V |
| $V_I$ (2) | Input voltage range | | -0.5 | $V_{CC}$+0.5 | V |
| $V_O$ (2) | Output voltage range | | -0.5 | $V_{CC}$+0.5 | V |
| $I_{IK}$ | Input clamp current | ($V_I$ < 0 or $V_I$ > $V_{CC}$) | | ±20 | mA |
| $I_{OK}$ | Output clamp current | ($V_O$ < 0 or $V_O$ > $V_{CC}$) | | ±20 | mA |
| $I_O$ | Continuous output current | ($V_O$ = 0 or $V_{CC}$) | | ±50 | mA |
| | Continuous current through $V_{CC}$ or GND | | | ±200 | mA |
| $T_{stg}$ | Storage temperature range | | –65 | 150 | °C |

(1) Stresses beyond those listed under “absolute maximum ratings” may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated under “recommended operating conditions” is not implied. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.
(2) The input and output voltage ratings may be exceeded if the input and output current ratings are observed.

## 4.2 Recommended Operating Conditions

over operating free-air temperature range (unless otherwise noted)(1)

| | | SN54ACT241 | | SN74ACT241 | | UNIT |
|---|---|---|---|---|---|---|
| | | MIN | MAX | MIN | MAX | |
| $V_{CC}$ | Supply voltage | 4.5 | 5.5 | 4.5 | 5.5 | V |
| $V_{IH}$ | High-level input voltage | 2 | | 2 | | V |
| $V_{IL}$ | Low-level input voltage | | 0.8 | | 0.8 | V |
| $V_I$ | Input voltage | 0 | $V_{CC}$ | 0 | $V_{CC}$ | V |
| $V_O$ | Output voltage | 0 | $V_{CC}$ | 0 | $V_{CC}$ | V |
| $I_{OH}$ | High-level output current | | -24 | | -24 | mA |
| $I_{OL}$ | Low-level output current | | 24 | | 24 | mA |
| Δt/Δv | Input transition rise or fall rate | | 8 | | 8 | ns/V |
| $T_A$ | Operating free-air temperature | –55 | 125 | –40 | 85 | °C |

(1) All unused inputs of the device must be held at $V_{CC}$ or GND to ensure proper device operation. Refer to the TI application report, *Implications of Slow or Floating CMOS Inputs*, literature number SCBA004.

## 4.3 Thermal Information

| THERMAL METRIC(1) | | DB (SSOP) | DW (SOIC) | N (PDIP) | NS (SOP) | PW (TSSOP) | UNIT |
|---|---|---|---|---|---|---|---|
| | | 20 PINS | | | | | |
| $R_{\theta JA}$ | Junction-to-ambient thermal resistance(2) | 70 | 58 | 69 | 60 | 126.2 | °C/W |

(1) For more information about traditional and new thermal metrics, see the *IC Package Thermal Metrics* application report, SPRA953.
(2) The package thermal impedance is calculated in accordance with JESD 51-7.


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## 4.4 Electrical Characteristics

over recommended operating free-air temperature range (unless otherwise noted)

| PARAMETER | TEST CONDITIONS | | $V_{CC}$ | $T_A$= 25°C | | | SN54ACT241 | | SN74ACT241 | | UNIT |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | MIN | TYP | MAX | MIN | MAX | MIN | MAX | |
| $V_{OH}$ | $I_{OH}$ = –50 µA | | 4.5 V | 4.4 | 4.49 | | 4.4 | | 4.4 | | V |
| | | | 5.5 V | 5.4 | 5.49 | | 5.4 | | 5.4 | | |
| | $I_{OH}$ = –24 mA | | 4.5 V | 3.86 | | | 3.7 | | 3.76 | | |
| | | | 5.5 V | 4.86 | | | 4.7 | | 4.76 | | |
| | $I_{OH}$ = –50 mA(1) | | 5.5 V | | | | 3.85 | | | | |
| | $I_{OH}$ = –75 mA(1) | | 5.5 V | | | | | | 3.85 | | |
| $V_{OL}$ | $I_{OL}$ = 50 µA | | 4.5 V | | 0.001 | 0.1 | | 0.1 | | 0.1 | V |
| | | | 5.5 V | | 0.001 | 0.1 | | 0.1 | | 0.1 | |
| | $I_{OL}$ = 24 mA | | 4.5 V | | | 0.36 | | 0.5 | | 0.44 | |
| | | | 5.5 V | | | 0.36 | | 0.5 | | 0.44 | |
| | $I_{OL}$ = 50 mA(1) | | 5.5 V | | | | | 1.65 | | | |
| | $I_{OL}$ = 75 mA(1) | | 5.5 V | | | | | | | 1.65 | |
| $I_{OZ}$ | $V_O$ = $V_{CC}$ or GND | | 5.5 V | | | ±0.25 | | ±5 | | ±2.5 | µA |
| $I_I$ | $V_I$ = $V_{CC}$ or GND | | 5.5 V | | | ±0.1 | | ±1 | | ±1 | µA |
| $I_{CC}$ | $V_I$ = $V_{CC}$ or GND, | $I_O$ = 0 | 5.5 V | | | 4 | | 80 | | 40 | µA |
| $\Delta I_{CC}$ (2) | One input at 3.4 V, Other inputs at GND or $V_{CC}$ | | 5.5 V | | 0.6 | | | 1.6 | | 1.5 | mA |
| $C_i$ | $V_I$ = $V_{CC}$ or GND | | 5 V | | 2.5 | | | | | | pF |
| $C_o$ | $V_I$ = $V_{CC}$ or GND | | 5 V | | 8 | | | | | | pF |

(1) Not more than one output should be tested at a time, and the duration of the test should not exceed 2 ms.
(2) This is the increase in supply current for each input that is at one of the specified TTL voltage levels, rather than 0 V or $V_{CC}$.

## 4.5 Switching Characteristics

over recommended operating free-air temperature range, $V_{CC}$ = 5 V ± 0.5 V (unless otherwise noted) (see Load Circuit and Voltage Waveforms)

| PARAMETER | FROM (INPUT) | TO (OUTPUT) | $T_A$ = 25°C | | | SN54ACT241 | | SN74ACT241 | | UNIT |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | MIN | TYP | MAX | MIN | MAX | MIN | MAX | |
| $t_{PLH}$ | A | Y | 1.5 | 6 | 8.5 | 1 | 9.5 | 1.5 | 9.5 | ns |
| $t_{PHL}$ | | | 1.5 | 5.5 | 7.5 | 1 | 9 | 1.5 | 8.5 | |
| $t_{PZH}$ | $\overline{OE}$ or OE | Y | 1.5 | 7 | 8.5 | 1 | 10 | 1 | 9.5 | ns |
| $t_{PZL}$ | | | 2 | 7 | 9.5 | 1 | 11.5 | 1.5 | 10.5 | |
| $t_{PHZ}$ | $\overline{OE}$ or OE | Y | 2 | 8 | 9.5 | 1 | 11 | 2 | 10.5 | ns |
| $t_{PLZ}$ | | | 2.5 | 6.5 | 10 | 1 | 11.5 | 2 | 10.5 | |

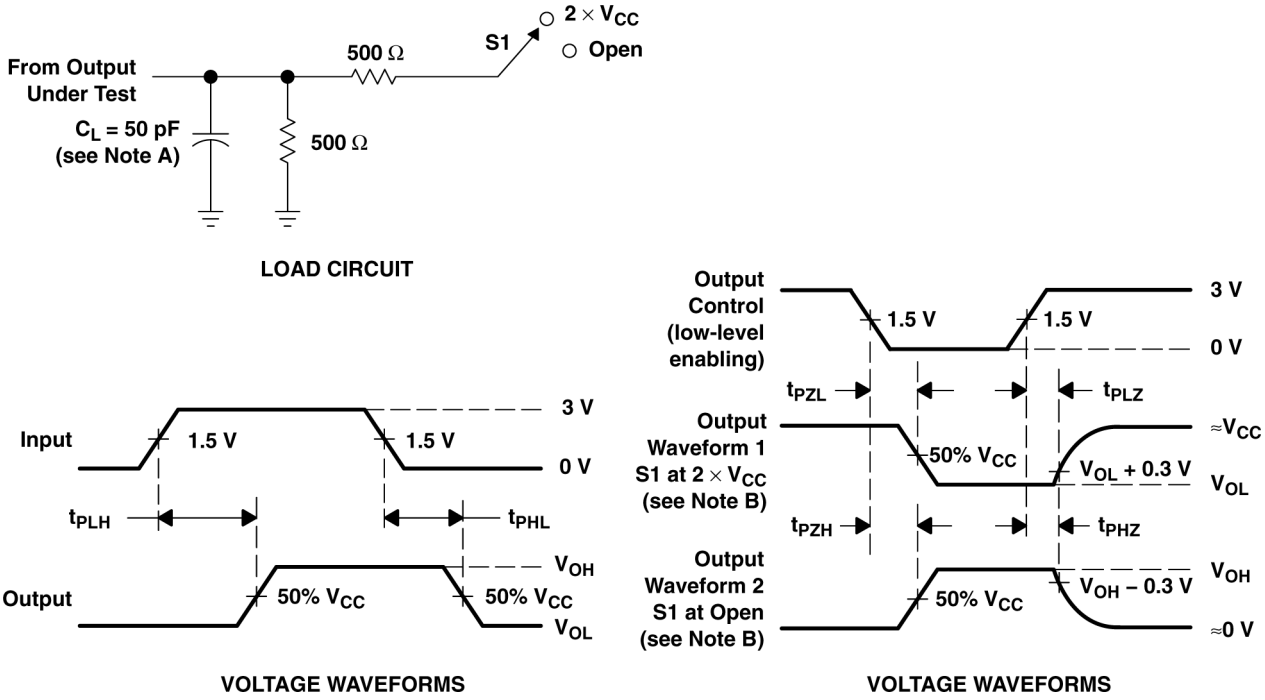
## 4.6 Operating Characteristics

$V_{CC}$ = 5 V, $T_A$ = 25°C

| PARAMETER | | TEST CONDITIONS | | TYP | UNIT |
|---|---|---|---|---|---|
| $C_{pd}$ | Power dissipation capacitance per buffer/driver | $C_L$ = 50 pF, | f = 1 MHz | 45 | pF |

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## 5 Parameter Measurement Information




A. $C_L$ includes probe and jig capacitance.
B. Waveform 1 is for an output with internal conditions such that the output is low except when disabled by the output control. Waveform 2 is for an output with internal conditions such that the output is high except when disabled by the output control.
C. All input pulses are supplied by generators having the following characteristics: PRR ≤ 1 MHz, $Z_O$ = 50 Ω, $t_r$ ≤ 2.5 ns, $t_f$ ≤ 2.5 ns.
D. The outputs are measured one at a time with one input transition per measurement.

**Figure 5-1. Load Circuit and Voltage Waveforms**

| TEST | S1 |
|---|---|
| $t_{PLH}/t_{PHL}$ | Open |
| $t_{PLZ}/t_{PZL}$ | 2 × $V_{CC}$ |
| $t_{PHZ}/t_{PZH}$ | Open |


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# 6 Detailed Description

## 6.1 Overview

The 'ACT241 devices are organized as two 4-bit buffers/drivers with separate complementary output-enable (1$\overline{OE}$ and 2OE) inputs. When 1$\overline{OE}$ is low or 2OE is high, the device passes noninverted data from the A inputs to the Y outputs. When 1$\overline{OE}$ is high or 2OE is low, the outputs are in the high-impedance state

To ensure the high-impedance state during power up or power down, $\overline{OE}$ should be tied to $V_{CC}$ through a pullup resistor and OE should be tied to GND through a pulldown resistor; the minimum value of the resistor is determined by the current-sinking or the current-sourcing capability of the driver.

## 6.2 Functional Block Diagram

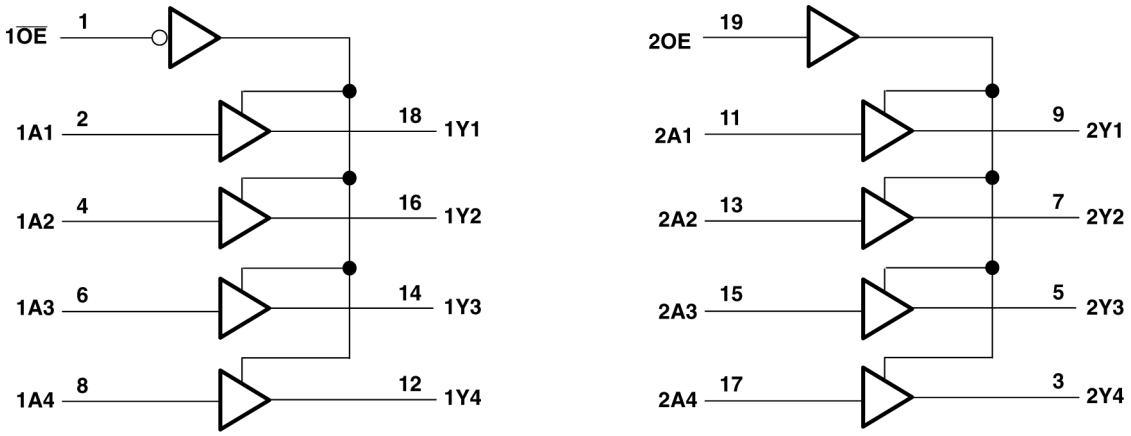



**Figure 6-1. Logic Diagram (Positive Logic)**

## 6.3 Device Functional Modes

### Function Tables

| INPUTS | | OUTPUT |
|---|---|---|
| 1$\overline{OE}$ | 1A | 1Y |
| L | H | H |
| L | L | L |
| H | X | Z |

| INPUTS | | OUTPUT |
|---|---|---|
| 2OE | 2A | 2Y |
| H | H | H |
| H | L | L |
| L | X | Z |

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# 7 Application and Implementation

**Note**

Information in the following applications sections is not part of the TI component specification, and TI does not warrant its accuracy or completeness. TI's customers are responsible for determining suitability of components for their purposes, as well as validating and testing their design implementation to confirm system functionality.

## 7.1 Power Supply Recommendations

The power supply can be any voltage between the min and max supply voltage rating located in *Section 4.2*.

Each $V_{CC}$ terminal should have a good bypass capacitor to prevent power disturbance. For devices with a single supply, TI recommends 0.1 µF and if there are multiple $V_{CC}$ terminals, then TI recommends .01 µF or .022 µF for each power terminal. It is okay to parallel multiple bypass capacitors to reject different frequencies of noise. A 0.1 µF and 1 µF are commonly used in parallel. The bypass capacitor should be installed as close to the power terminal as possible for best results.

## 7.2 Layout

### 7.2.1 Layout Guidelines

When using multiple bit logic devices inputs should not ever float.

In many cases, functions or parts of functions of digital logic devices are unused, for example, when only two inputs of a triple-input AND gate are used or only 3 of the 4 buffer gates are used. Such input pins should not be left unconnected because the undefined voltages at the outside connections result in undefined operational states. Specified below are the rules that must be observed under all circumstances. All unused inputs of digital logic devices must be connected to a high or low bias to prevent them from floating. The logic level that should be applied to any particular unused input depends on the function of the device. Generally they will be tied to GND or $V_{CC}$ whichever make more sense or is more convenient. It is generally okay to float outputs unless the part is a transceiver. If the transceiver has an output enable pin it will disable the outputs section of the part when asserted. This does not disable the input section of the IOs so they cannot float when disabled.

#### *7.2.1.1 Layout Example*

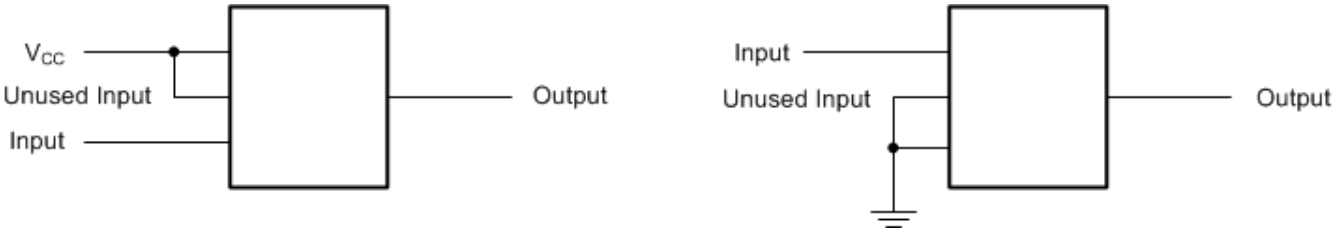



**Figure 7-1. Layout Recommendation**


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# 8 Device and Documentation Support

## 8.1 Documentation Support

### 8.1.1 Related Documentation

The table below lists quick access links. Categories include technical documents, support and community resources, tools and software, and quick access to sample or buy.

**Table 8-1. Related Links**

| PARTS | PRODUCT FOLDER | SAMPLE & BUY | TECHNICAL DOCUMENTS | TOOLS & SOFTWARE | SUPPORT & COMMUNITY |
|---|---|---|---|---|---|
| SN54ACT241 | Click here | Click here | Click here | Click here | Click here |
| SN74ACT241 | Click here | Click here | Click here | Click here | Click here |

## 8.2 Receiving Notification of Documentation Updates

To receive notification of documentation updates, navigate to the device product folder on ti.com. Click on *Notifications* to register and receive a weekly digest of any product information that has changed. For change details, review the revision history included in any revised document.

## 8.3 Support Resources

TI E2E™ support forums are an engineer's go-to source for fast, verified answers and design help — straight from the experts. Search existing answers or ask your own question to get the quick design help you need.

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## 8.4 Trademarks

TI E2E™ is a trademark of Texas Instruments.
All trademarks are the property of their respective owners.

## 8.5 Electrostatic Discharge Caution



This integrated circuit can be damaged by ESD. Texas Instruments recommends that all integrated circuits be handled with appropriate precautions. Failure to observe proper handling and installation procedures can cause damage.

ESD damage can range from subtle performance degradation to complete device failure. Precision integrated circuits may be more susceptible to damage because very small parametric changes could cause the device not to meet its published specifications.

## 8.6 Glossary

TI Glossary — This glossary lists and explains terms, acronyms, and definitions.

# 9 Revision History

| Changes from Revision D (May 2023) to Revision E (March 2024) | Page |
|---|---|
| • Added *Application and Implementation* section, *Device and Documentation Support* section, package size to *Device Information* table, and *Mechanical, Packaging, and Orderable Information* section | 1 |
| • Updated RθJA value: PW = 83 to 126.2, all values in °C/W | 4 |

| Changes from Revision C (October 2002) to Revision D (May 2023) | Page |
|---|---|
| • Added *Package Information* table, *Pin Functions* table, and *Thermal Information* table | 1 |

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## 10 Mechanical, Packaging, and Orderable Information

The following pages include mechanical, packaging, and orderable information. This information is the most current data available for the designated devices. This data is subject to change without notice and revision of this document. For browser-based versions of this data sheet, refer to the left-hand navigation.


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## PACKAGING INFORMATION

| Orderable part number | Status (1) | Material type (2) | Package \| Pins | Package qty \| Carrier | RoHS (3) | Lead finish/ Ball material (4) | MSL rating/ Peak reflow (5) | Op temp (°C) | Part marking (6) |
|---|---|---|---|---|---|---|---|---|---|
| 5962-89847012A | Active | Production | LCCC (FK) \| 20 | 55 \| TUBE | No | SNPB | N/A for Pkg Type | -55 to 125 | 5962-89847012A SNJ54ACT 241FK |
| 5962-8984701RA | Active | Production | CDIP (J) \| 20 | 20 \| TUBE | No | SNPB | N/A for Pkg Type | -55 to 125 | 5962-8984701RA SNJ54ACT241J |
| 5962-8984701SA | Active | Production | CFP (W) \| 20 | 25 \| TUBE | No | SNPB | N/A for Pkg Type | -55 to 125 | 5962-8984701SA SNJ54ACT241W |
| SN74ACT241DBR | Last Time Buy | Production | SSOP (DB) \| 20 | 2000 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -40 to 85 | AD241 |
| SN74ACT241DBR.A | Last Time Buy | Production | SSOP (DB) \| 20 | 2000 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -40 to 85 | AD241 |
| SN74ACT241DW | Obsolete | Production | SOIC (DW) \| 20 | - | - | Call TI | Call TI | -40 to 85 | ACT241 |
| SN74ACT241DWR | Active | Production | SOIC (DW) \| 20 | 2000 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -40 to 85 | ACT241 |
| SN74ACT241DWR.A | Active | Production | SOIC (DW) \| 20 | 2000 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -40 to 85 | ACT241 |
| SN74ACT241N | Active | Production | PDIP (N) \| 20 | 20 \| TUBE | Yes | NIPDAU | N/A for Pkg Type | -40 to 85 | SN74ACT241N |
| SN74ACT241N.A | Active | Production | PDIP (N) \| 20 | 20 \| TUBE | Yes | NIPDAU | N/A for Pkg Type | -40 to 85 | SN74ACT241N |
| SN74ACT241NSR | Active | Production | SOP (NS) \| 20 | 2000 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -40 to 85 | ACT241 |
| SN74ACT241NSR.A | Active | Production | SOP (NS) \| 20 | 2000 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -40 to 85 | ACT241 |
| SN74ACT241PW | Obsolete | Production | TSSOP (PW) \| 20 | - | - | Call TI | Call TI | -40 to 85 | AD241 |
| SN74ACT241PWR | Active | Production | TSSOP (PW) \| 20 | 2000 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -40 to 85 | AD241 |
| SN74ACT241PWR.A | Active | Production | TSSOP (PW) \| 20 | 2000 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -40 to 85 | AD241 |
| SNJ54ACT241FK | Active | Production | LCCC (FK) \| 20 | 55 \| TUBE | No | SNPB | N/A for Pkg Type | -55 to 125 | 5962-89847012A SNJ54ACT 241FK |
| SNJ54ACT241FK.A | Active | Production | LCCC (FK) \| 20 | 55 \| TUBE | No | SNPB | N/A for Pkg Type | -55 to 125 | 5962-89847012A SNJ54ACT 241FK |
| SNJ54ACT241J | Active | Production | CDIP (J) \| 20 | 20 \| TUBE | No | SNPB | N/A for Pkg Type | -55 to 125 | 5962-8984701RA SNJ54ACT241J |

| Orderable part number | Status (1) | Material type (2) | Package \| Pins | Package qty \| Carrier | RoHS (3) | Lead finish/ Ball material (4) | MSL rating/ Peak reflow (5) | Op temp (°C) | Part marking (6) |
|---|---|---|---|---|---|---|---|---|---|
| SNJ54ACT241J.A | Active | Production | CDIP (J) \| 20 | 20 \| TUBE | No | SNPB | N/A for Pkg Type | -55 to 125 | 5962-8984701RA SNJ54ACT241J |
| SNJ54ACT241W | Active | Production | CFP (W) \| 20 | 25 \| TUBE | No | SNPB | N/A for Pkg Type | -55 to 125 | 5962-8984701SA SNJ54ACT241W |
| SNJ54ACT241W.A | Active | Production | CFP (W) \| 20 | 25 \| TUBE | No | SNPB | N/A for Pkg Type | -55 to 125 | 5962-8984701SA SNJ54ACT241W |

(1) **Status:** For more details on status, see our product life cycle.

(2) **Material type:** When designated, preproduction parts are prototypes/experimental devices, and are not yet approved or released for full production. Testing and final process, including without limitation quality assurance, reliability performance testing, and/or process qualification, may not yet be complete, and this item is subject to further changes or possible discontinuation. If available for ordering, purchases will be subject to an additional waiver at checkout, and are intended for early internal evaluation purposes only. These items are sold without warranties of any kind.

(3) **RoHS values:** Yes, No, RoHS Exempt. See the TI RoHS Statement for additional information and value definition.

(4) **Lead finish/Ball material:** Parts may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

(5) **MSL rating/Peak reflow:** The moisture sensitivity level ratings and peak solder (reflow) temperatures. In the event that a part has multiple moisture sensitivity ratings, only the lowest level per JEDEC standards is shown. Refer to the shipping label for the actual reflow temperature that will be used to mount the part to the printed circuit board.

(6) **Part marking:** There may be an additional marking, which relates to the logo, the lot trace code information, or the environmental category of the part.

Multiple part markings will be inside parentheses. Only one part marking contained in parentheses and separated by a "~" will appear on a part. If a line is indented then it is a continuation of the previous line and the two combined represent the entire part marking for that device.

**Important Information and Disclaimer:**The information provided on this page represents TI's knowledge and belief as of the date that it is provided. TI bases its knowledge and belief on information provided by third parties, and makes no representation or warranty as to the accuracy of such information. Efforts are underway to better integrate information from third parties. TI has taken and continues to take reasonable steps to provide representative and accurate information but may not have conducted destructive testing or chemical analysis on incoming materials and chemicals. TI and TI suppliers consider certain information to be proprietary, and thus CAS numbers and other limited information may not be available for release.

In no event shall TI's liability arising out of such information exceed the total purchase price of the TI part(s) at issue in this document sold by TI to Customer on an annual basis.

**OTHER QUALIFIED VERSIONS OF SN54ACT241, SN74ACT241 :**

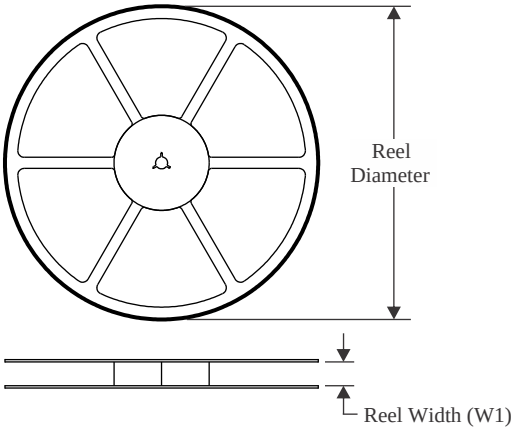
- Catalog : SN74ACT241
- Military : SN54ACT241

NOTE: Qualified Version Definitions:

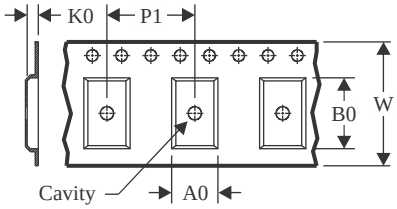
- Catalog - TI's standard catalog product
- Military - QML certified for Military and Defense Applications

## TAPE AND REEL INFORMATION

**REEL DIMENSIONS**

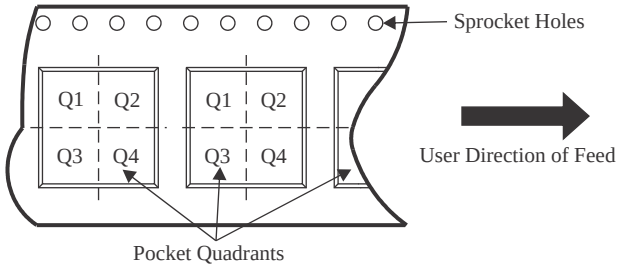



**TAPE DIMENSIONS**




| | |
|---|---|
| A0 | Dimension designed to accommodate the component width |
| B0 | Dimension designed to accommodate the component length |
| K0 | Dimension designed to accommodate the component thickness |
| W | Overall width of the carrier tape |
| P1 | Pitch between successive cavity centers |

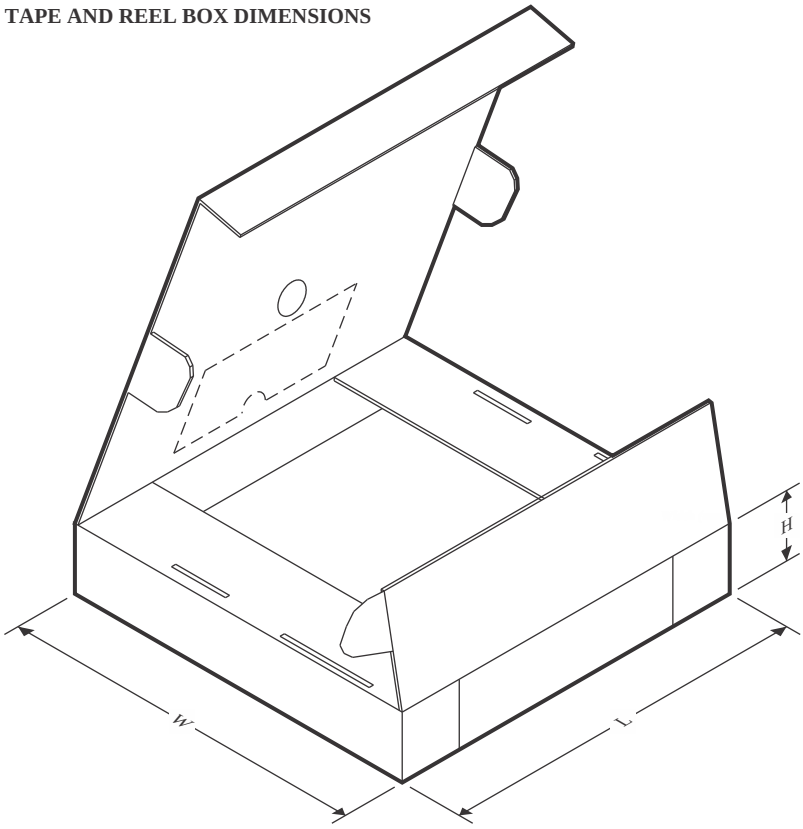
**QUADRANT ASSIGNMENTS FOR PIN 1 ORIENTATION IN TAPE**




*All dimensions are nominal

| Device | Package Type | Package Drawing | Pins | SPQ | Reel Diameter (mm) | Reel Width W1 (mm) | A0 (mm) | B0 (mm) | K0 (mm) | P1 (mm) | W (mm) | Pin1 Quadrant |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| SN74ACT241DBR | SSOP | DB | 20 | 2000 | 330.0 | 16.4 | 8.2 | 7.5 | 2.5 | 12.0 | 16.0 | Q1 |
| SN74ACT241DWR | SOIC | DW | 20 | 2000 | 330.0 | 24.4 | 10.8 | 13.3 | 2.7 | 12.0 | 24.0 | Q1 |
| SN74ACT241NSR | SOP | NS | 20 | 2000 | 330.0 | 24.4 | 8.4 | 13.0 | 2.5 | 12.0 | 24.0 | Q1 |
| SN74ACT241PWR | TSSOP | PW | 20 | 2000 | 330.0 | 16.4 | 6.95 | 7.0 | 1.4 | 8.0 | 16.0 | Q1 |

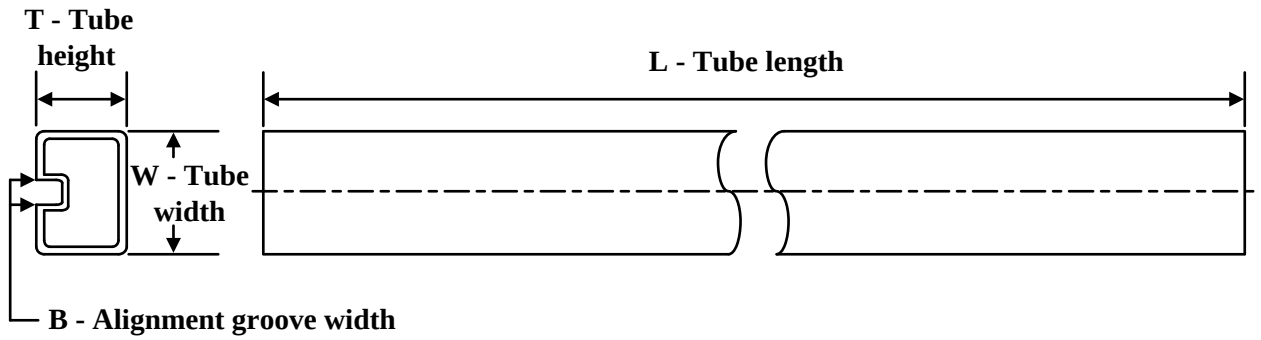
TAPE AND REEL BOX DIMENSIONS




*All dimensions are nominal

| Device | Package Type | Package Drawing | Pins | SPQ | Length (mm) | Width (mm) | Height (mm) |
|---|---|---|---|---|---|---|---|
| SN74ACT241DBR | SSOP | DB | 20 | 2000 | 353.0 | 353.0 | 32.0 |
| SN74ACT241DWR | SOIC | DW | 20 | 2000 | 356.0 | 356.0 | 45.0 |
| SN74ACT241NSR | SOP | NS | 20 | 2000 | 356.0 | 356.0 | 45.0 |
| SN74ACT241PWR | TSSOP | PW | 20 | 2000 | 353.0 | 353.0 | 32.0 |

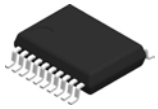
## TUBE




*All dimensions are nominal

| Device | Package Name | Package Type | Pins | SPQ | L (mm) | W (mm) | T (µm) | B (mm) |
|---|---|---|---|---|---|---|---|---|
| 5962-89847012A | FK | LCCC | 20 | 55 | 506.98 | 12.06 | 2030 | NA |
| 5962-8984701SA | W | CFP | 20 | 25 | 506.98 | 26.16 | 6220 | NA |
| SN74ACT241N | N | PDIP | 20 | 20 | 506 | 13.97 | 11230 | 4.32 |
| SN74ACT241N.A | N | PDIP | 20 | 20 | 506 | 13.97 | 11230 | 4.32 |
| SNJ54ACT241FK | FK | LCCC | 20 | 55 | 506.98 | 12.06 | 2030 | NA |
| SNJ54ACT241FK.A | FK | LCCC | 20 | 55 | 506.98 | 12.06 | 2030 | NA |
| SNJ54ACT241W | W | CFP | 20 | 25 | 506.98 | 26.16 | 6220 | NA |
| SNJ54ACT241W.A | W | CFP | 20 | 25 | 506.98 | 26.16 | 6220 | NA |

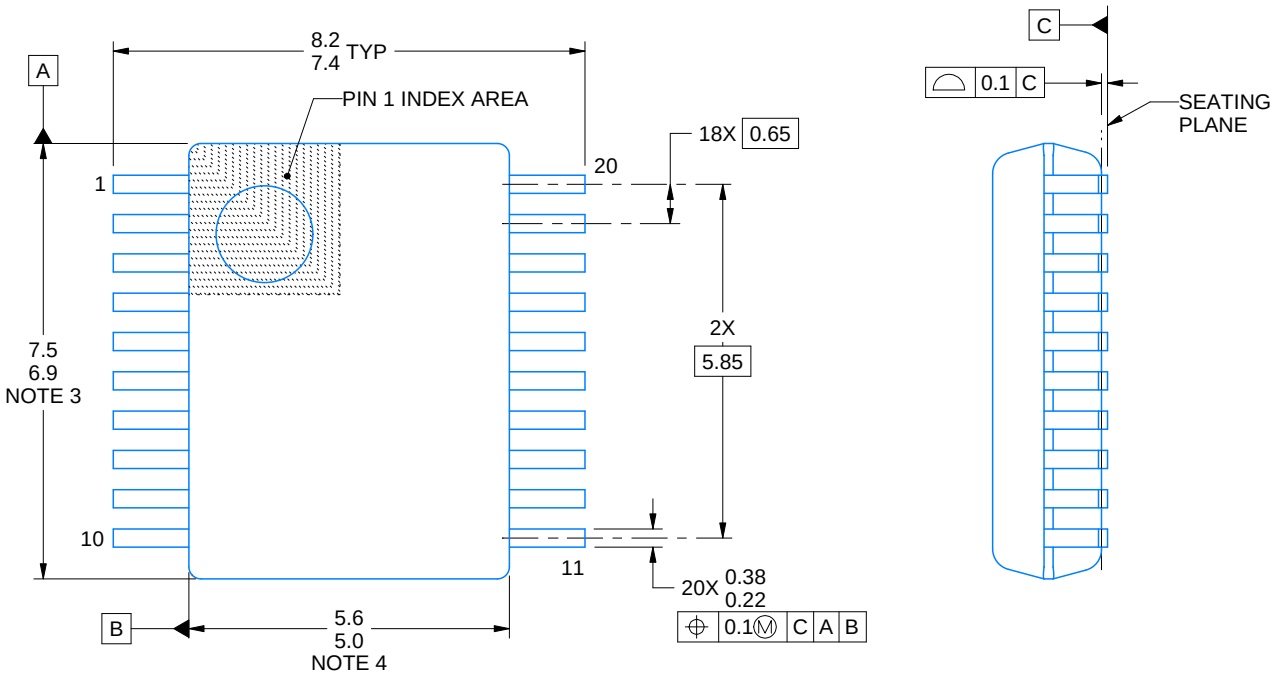
# DB0020A

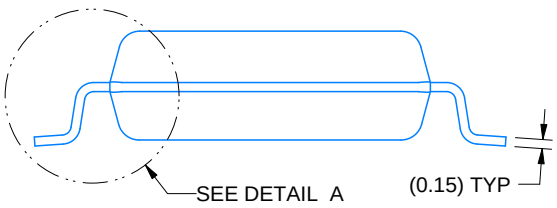


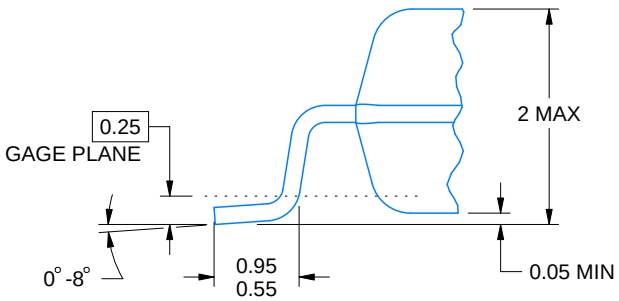
# PACKAGE OUTLINE

## SSOP - 2 mm max height

SMALL OUTLINE PACKAGE



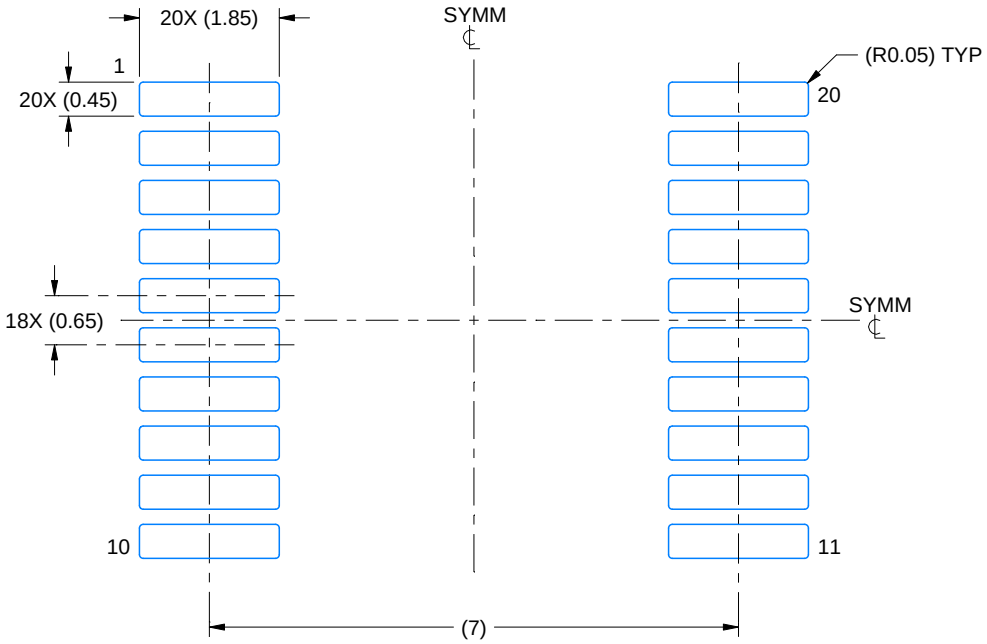




DETAIL A
TYPICAL

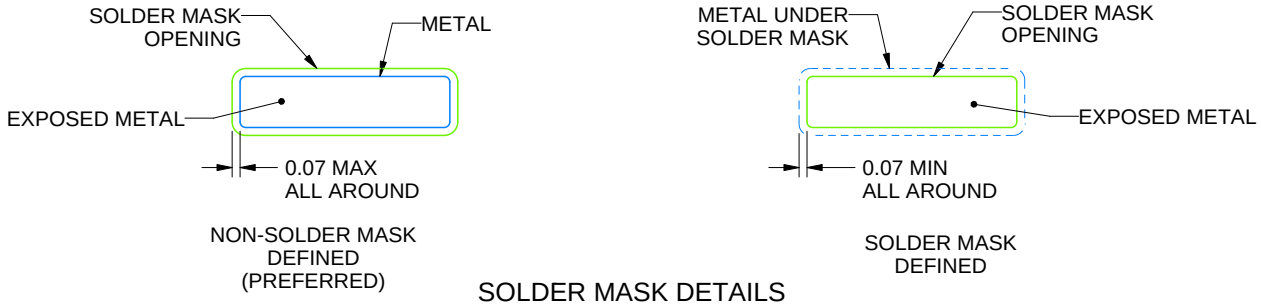
4214851/B 08/2019

NOTES:

1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm per side.
4. This dimension does not include interlead flash. Interlead flash shall not exceed 0.25 mm per side.
5. Reference JEDEC registration MO-150.




LAND PATTERN EXAMPLE
EXPOSED METAL SHOWN
SCALE: 10X




SOLDER MASK DETAILS

4214851/B 08/2019

NOTES: (continued)

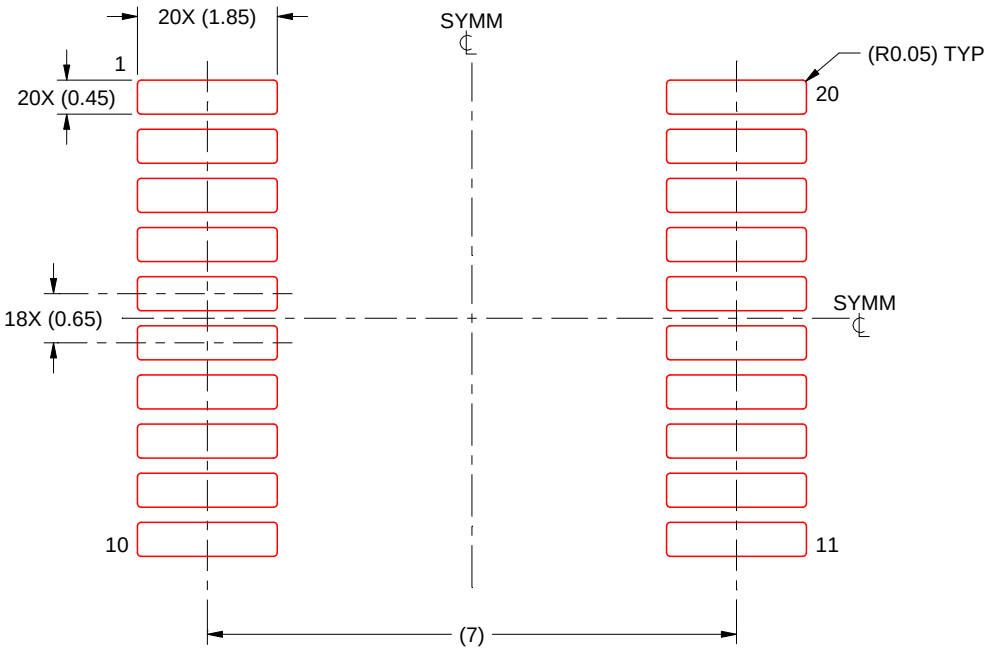
6. Publication IPC-7351 may have alternate designs.
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.

# EXAMPLE STENCIL DESIGN

## DB0020A

## SSOP - 2 mm max height

SMALL OUTLINE PACKAGE




SOLDER PASTE EXAMPLE
BASED ON 0.125 mm THICK STENCIL
SCALE: 10X

4214851/B 08/2019

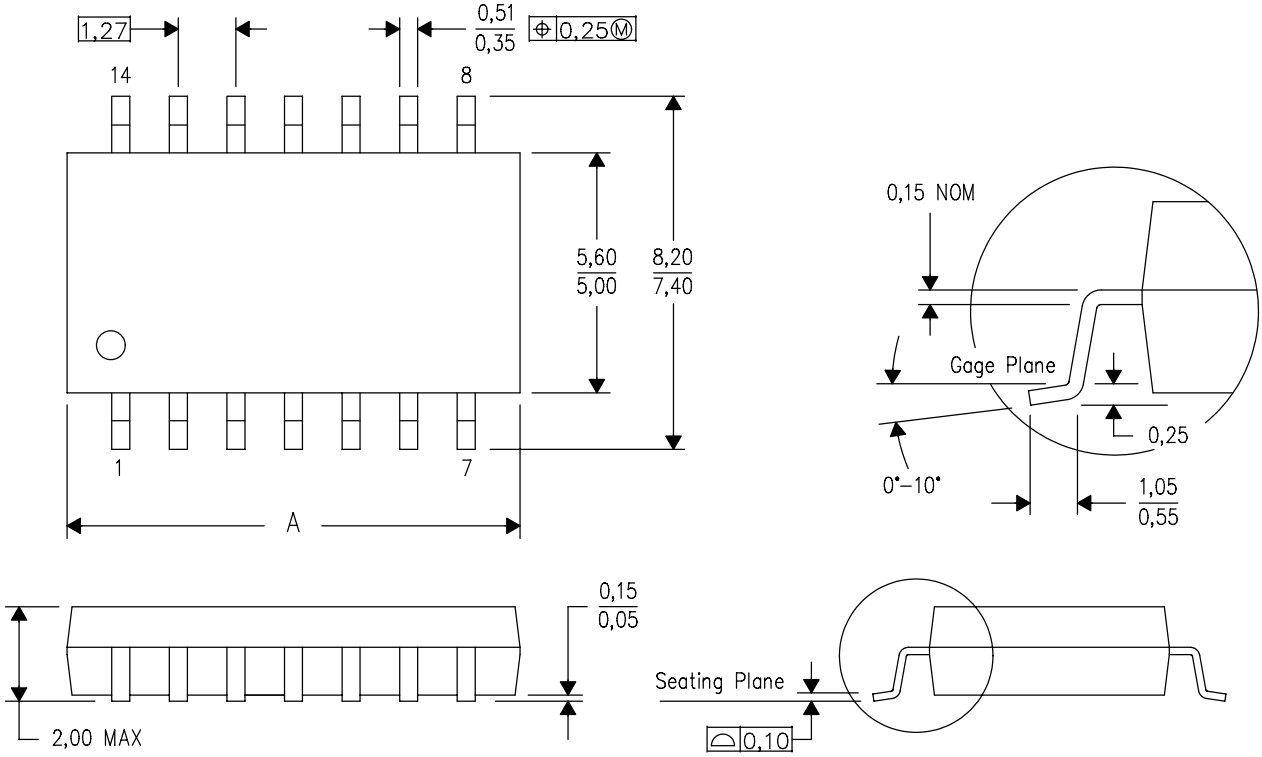
NOTES: (continued)

8. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
9. Board assembly site may have different recommendations for stencil design.

## MECHANICAL DATA

**NS (R-PDSO-G**)** **PLASTIC SMALL-OUTLINE PACKAGE**

**14-PINS SHOWN**




| DIM \ PINS ** | 14 | 16 | 20 | 24 |
|---|---|---|---|---|
| A MAX | 10,50 | 10,50 | 12,90 | 15,30 |
| A MIN | 9,90 | 9,90 | 12,30 | 14,70 |

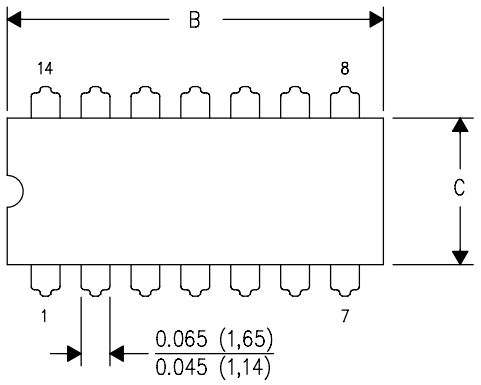
4040062/C 03/03

NOTES:
A. All linear dimensions are in millimeters.
B. This drawing is subject to change without notice.
C. Body dimensions do not include mold flash or protrusion, not to exceed 0,15.

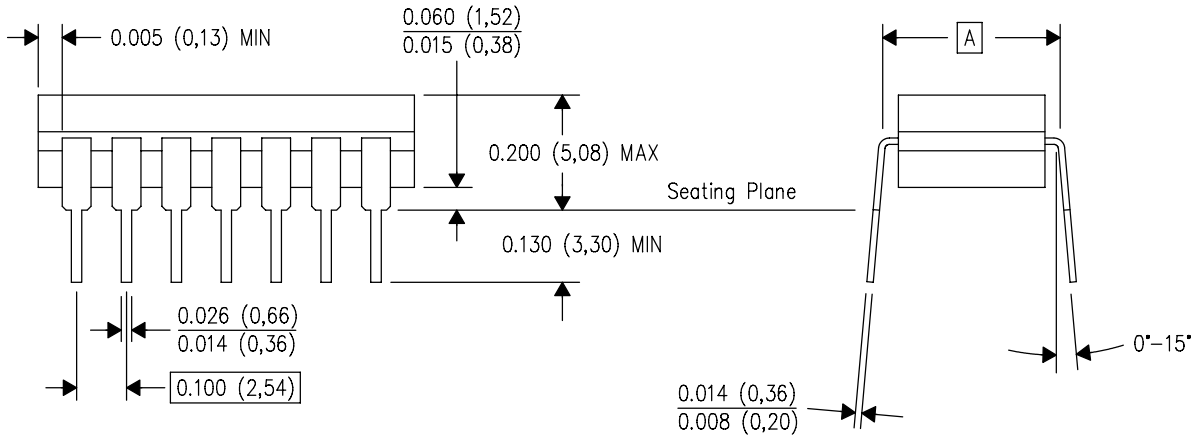
J (R–GDIP–T**)

14 LEADS SHOWN

# CERAMIC DUAL IN–LINE PACKAGE




| DIM \ PINS ** | 14 | 16 | 18 | 20 |
|---|---|---|---|---|
| A | 0.300 (7,62) BSC | 0.300 (7,62) BSC | 0.300 (7,62) BSC | 0.300 (7,62) BSC |
| B MAX | 0.785 (19,94) | .840 (21,34) | 0.960 (24,38) | 1.060 (26,92) |
| B MIN | — | — | — | — |
| C MAX | 0.300 (7,62) | 0.300 (7,62) | 0.310 (7,87) | 0.300 (7,62) |
| C MIN | 0.245 (6,22) | 0.245 (6,22) | 0.220 (5,59) | 0.245 (6,22) |




4040083/F 03/03

NOTES:
A. All linear dimensions are in inches (millimeters).
B. This drawing is subject to change without notice.
C. This package is hermetically sealed with a ceramic lid using glass frit.
D. Index point is provided on cap for terminal identification only on press ceramic glass frit seal only.
E. Falls within MIL STD 1835 GDIP1–T14, GDIP1–T16, GDIP1–T18 and GDIP1–T20.

# GENERIC PACKAGE VIEW

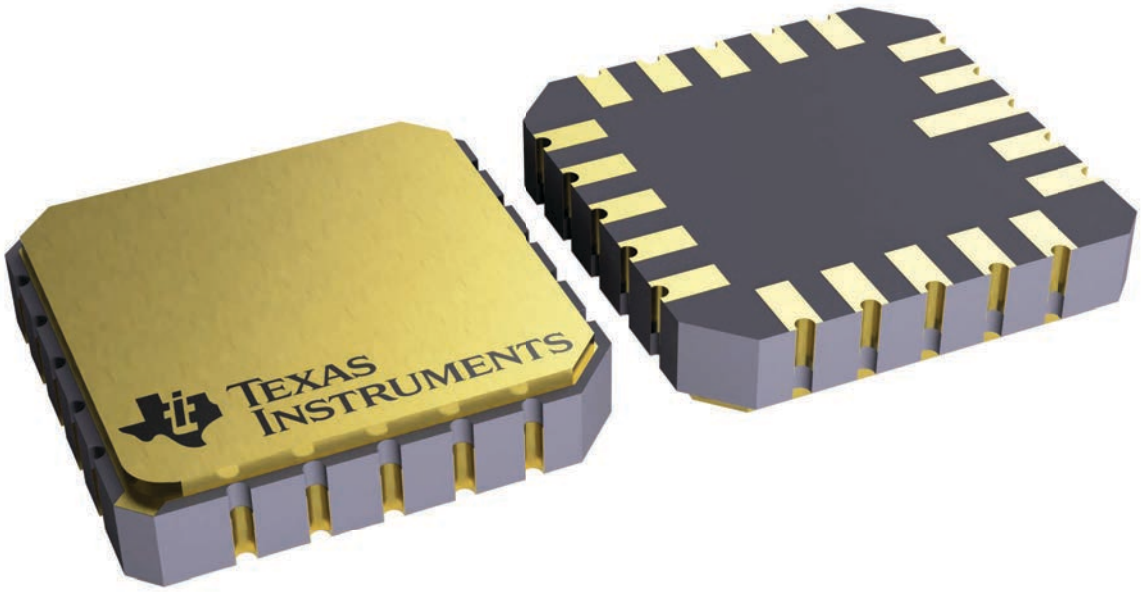
## FK 20

## LCCC - 2.03 mm max height

**8.89 x 8.89, 1.27 mm pitch**

LEADLESS CERAMIC CHIP CARRIER

This image is a representation of the package family, actual package may vary.
Refer to the product data sheet for package details.

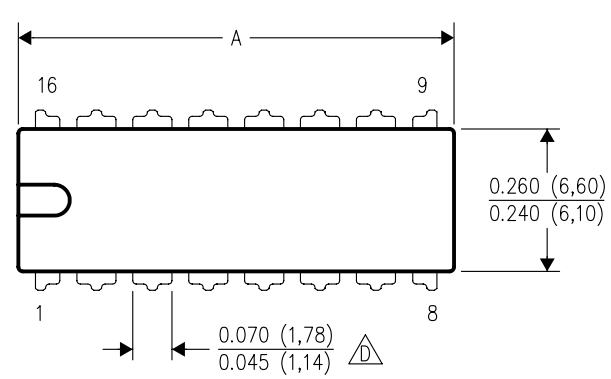



4229370VA\

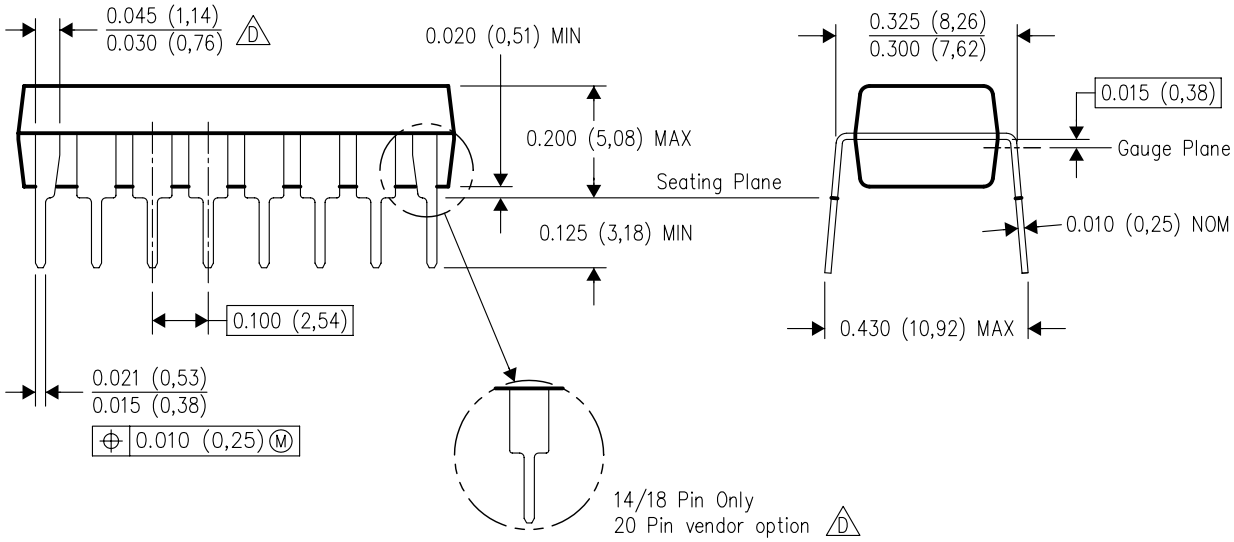
## N (R–PDIP–T**)

## PLASTIC DUAL–IN–LINE PACKAGE

16 PINS SHOWN

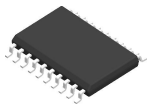



| DIM \ PINS ** | 14 | 16 | 18 | 20 |
|---|---|---|---|---|
| A MAX | 0.775 (19,69) | 0.775 (19,69) | 0.920 (23,37) | 1.060 (26,92) |
| A MIN | 0.745 (18,92) | 0.745 (18,92) | 0.850 (21,59) | 0.940 (23,88) |
| MS–001 VARIATION (C) | AA | BB | AC | AD |




4040049/E 12/2002

NOTES:
A. All linear dimensions are in inches (millimeters).
B. This drawing is subject to change without notice.
C. Falls within JEDEC MS–001, except 18 and 20 pin minimum body length (Dim A).
D. The 20 pin end lead shoulder width is a vendor option, either half or full width.

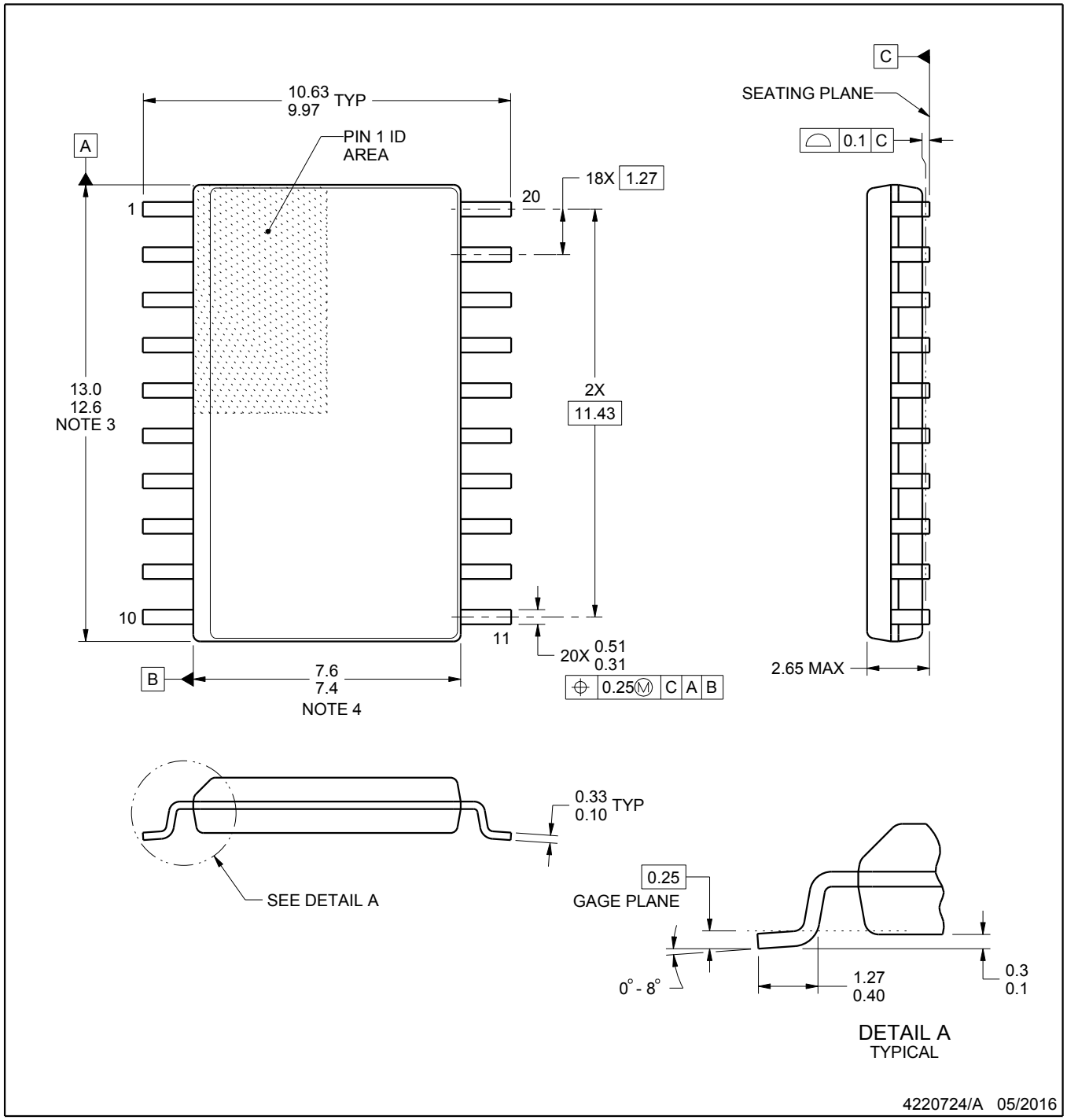


# PACKAGE OUTLINE

## DW0020A

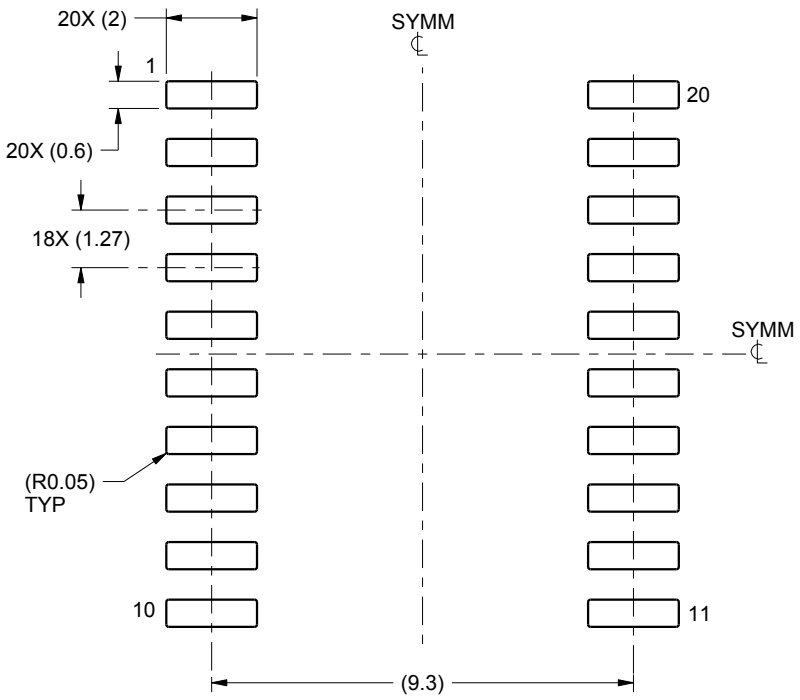
## SOIC - 2.65 mm max height

SOIC

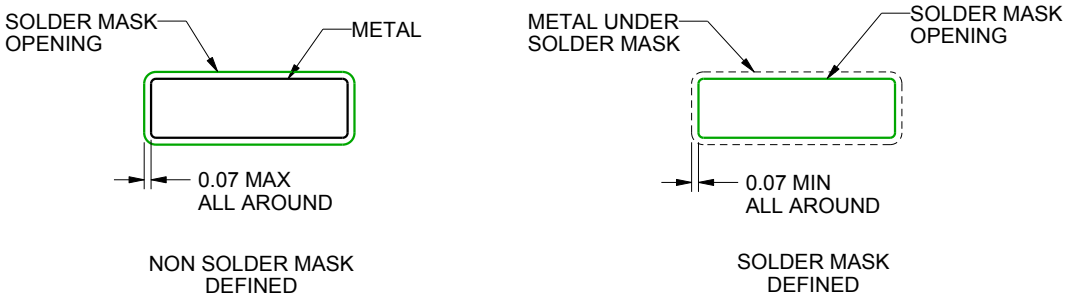



NOTES:

1. All linear dimensions are in millimeters. Dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm per side.
4. This dimension does not include interlead flash. Interlead flash shall not exceed 0.43 mm per side.
5. Reference JEDEC registration MS-013.




LAND PATTERN EXAMPLE
SCALE:6X

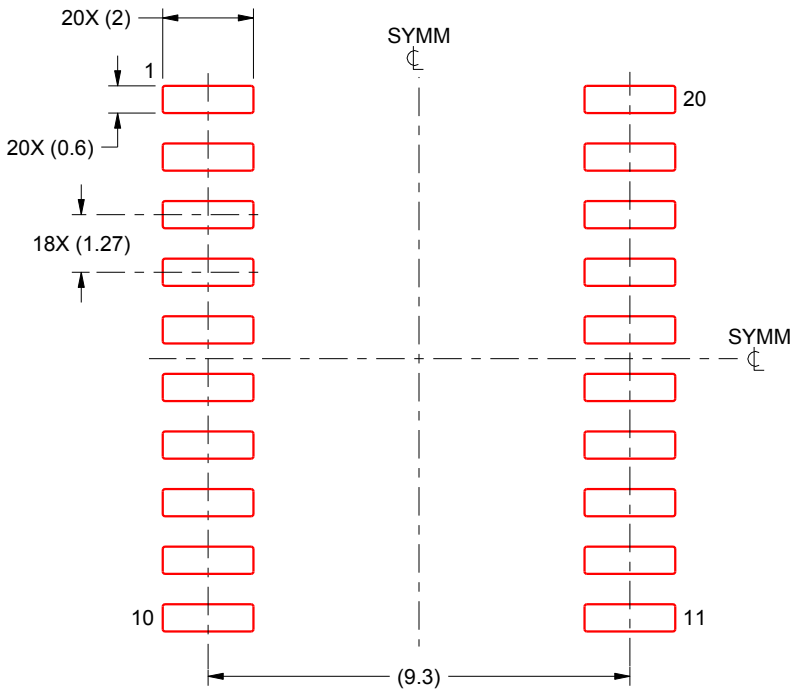



SOLDER MASK DETAILS

4220724/A 05/2016

NOTES: (continued)

6. Publication IPC-7351 may have alternate designs.
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.



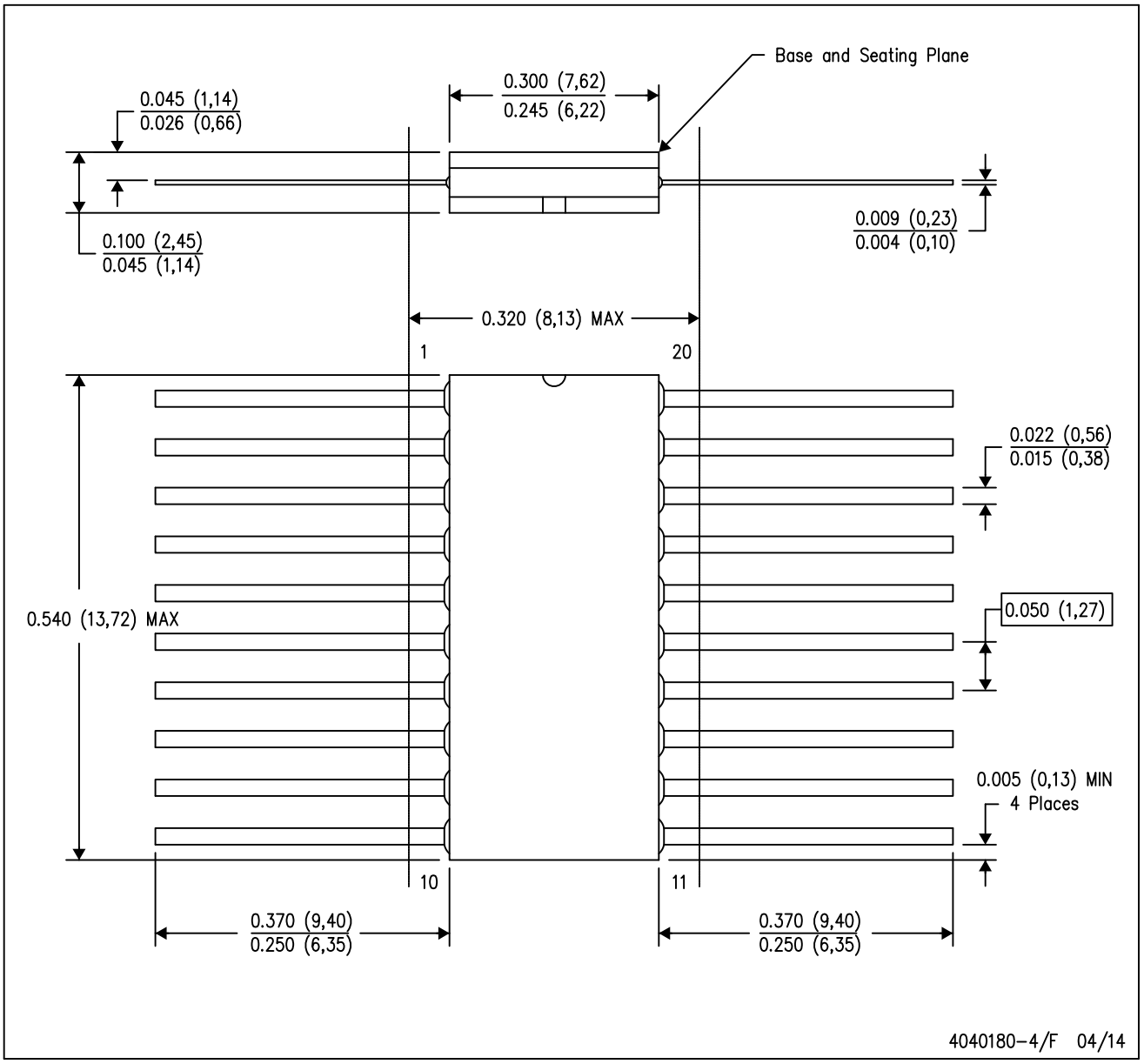

SOLDER PASTE EXAMPLE
BASED ON 0.125 mm THICK STENCIL
SCALE:6X

4220724/A 05/2016

NOTES: (continued)

8. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
9. Board assembly site may have different recommendations for stencil design.

W (R–GDFP–F20) CERAMIC DUAL FLATPACK



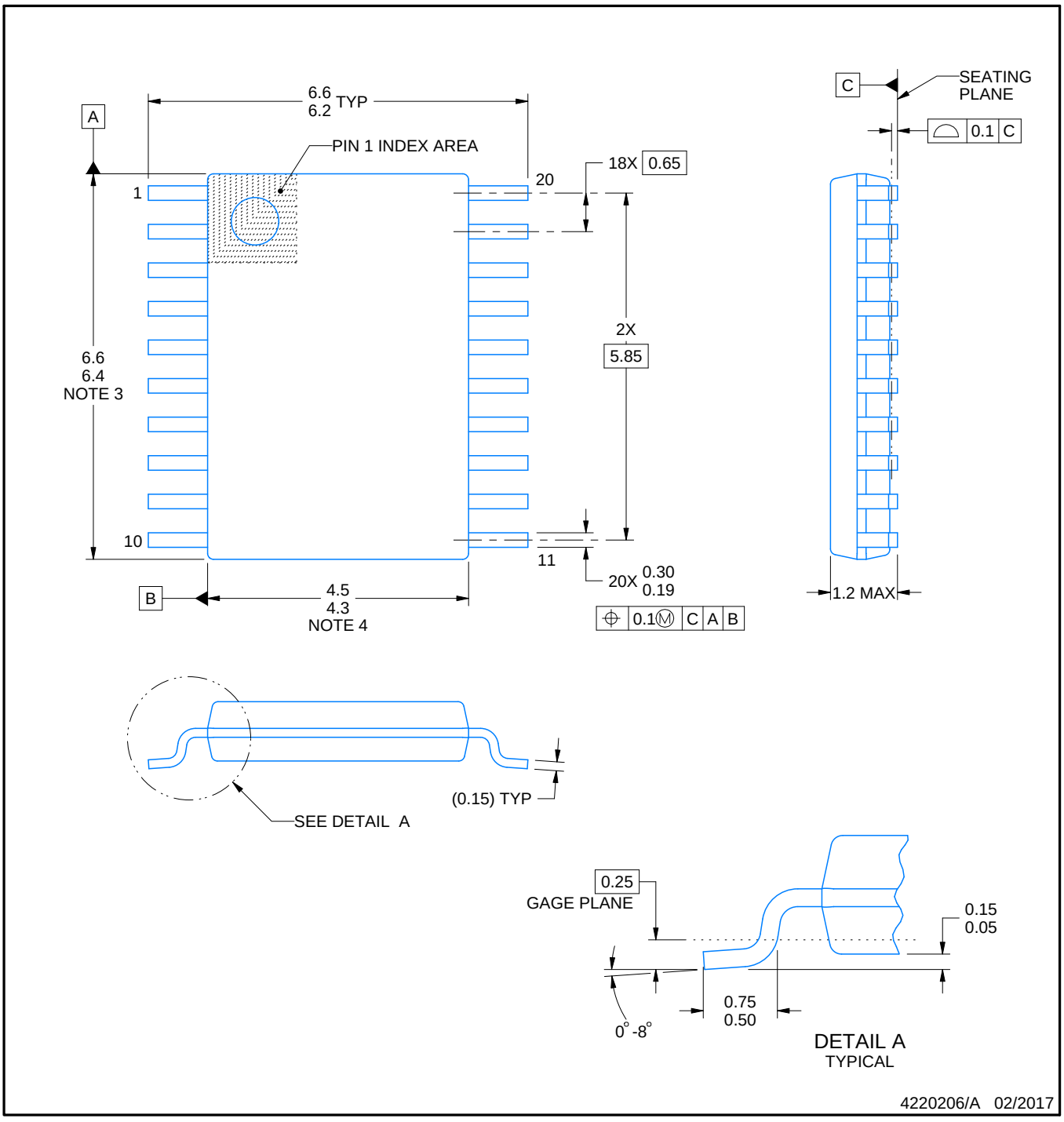

NOTES:
A. All linear dimensions are in inches (millimeters).
B. This drawing is subject to change without notice.
C. This package can be hermetically sealed with a ceramic lid using glass frit.
D. Index point is provided on cap for terminal identification only.
E. Falls within Mil–Std 1835 GDFP2–F20

# PACKAGE OUTLINE

## PW0020A

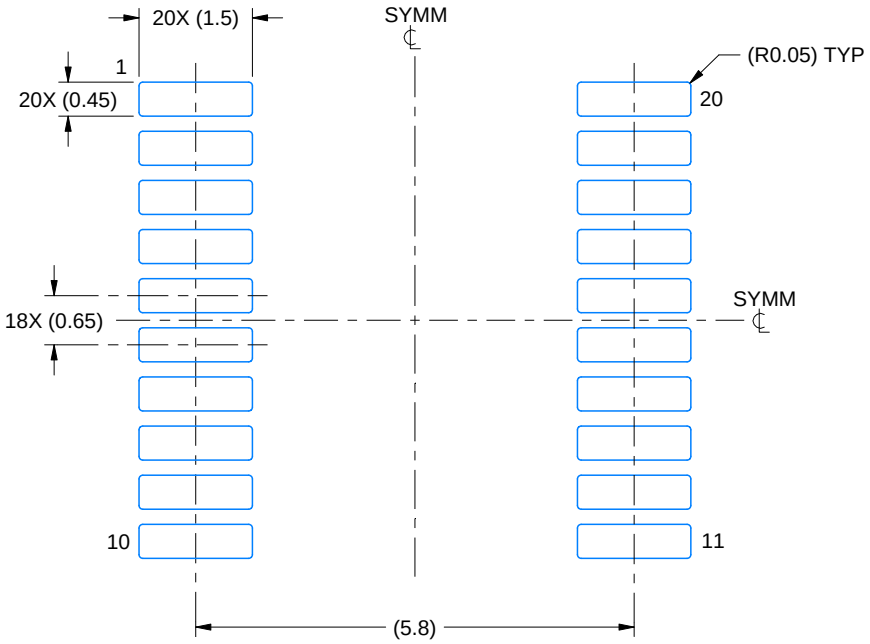
## TSSOP - 1.2 mm max height

SMALL OUTLINE PACKAGE




NOTES:

1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm per side.
4. This dimension does not include interlead flash. Interlead flash shall not exceed 0.25 mm per side.
5. Reference JEDEC registration MO-153.




LAND PATTERN EXAMPLE
EXPOSED METAL SHOWN
SCALE: 10X




NON-SOLDER MASK DEFINED (PREFERRED)

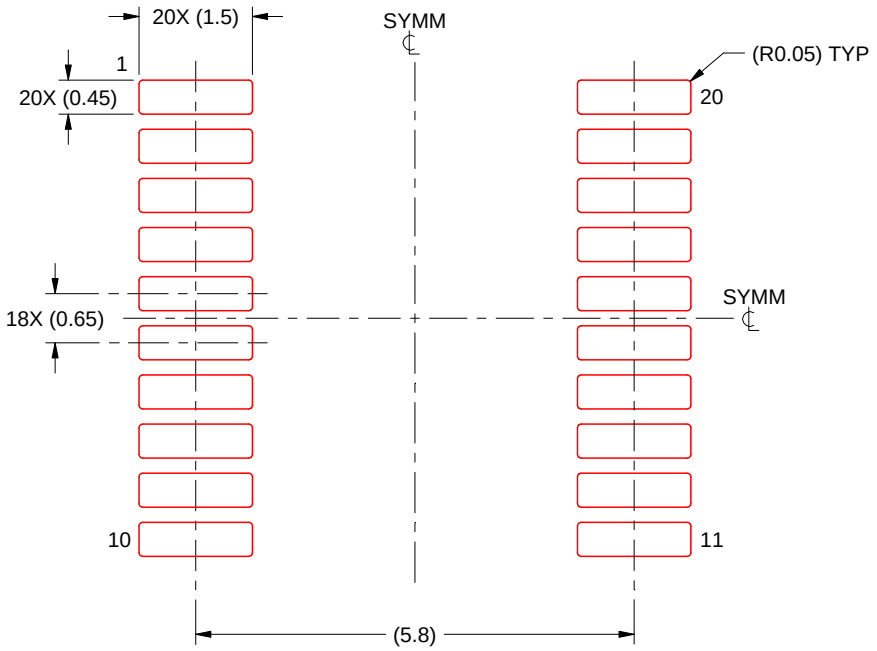
SOLDER MASK DEFINED

SOLDER MASK DETAILS

4220206/A 02/2017

NOTES: (continued)

6. Publication IPC-7351 may have alternate designs.
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.




SOLDER PASTE EXAMPLE
BASED ON 0.125 mm THICK STENCIL
SCALE: 10X

4220206/A 02/2017

NOTES: (continued)

8. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
9. Board assembly site may have different recommendations for stencil design.

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